

FACULDADE DE ENGENHARIA DA UNIVERSIDADE DO PORTO

Bridging the gap between forecasting accuracy and utility: case study with binary options

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Mestrado em Engenharia e Ciência de Dados

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Abstract

This thesis explores the discrepancy between predictive accuracy and real-world trading utility in binary options. Although statistical performance metrics like AUC, Accuracy, and F1 Score effectively measure a model's classification ability, they rarely capture the actual economic implications of such forecasts. Binary options, with their fixed all-or-nothing payouts, present a distinct opportunity to examine whether widely used forecasting metrics align with profitability in practice.

A range of machine learning models are trained and calibrated to produce probabilistic forecasts, which are subsequently evaluated not only by traditional statistical metrics but also by utility-based measures including Sharpe Ratio, Sortino Ratio and Average ROI per Trade. The empirical results demonstrate that statistical accuracy does not reliably translate into financial returns. Instead, model calibration and utility-aware evaluation frameworks are critical for practical success. These findings highlight the necessity of shifting focus from pure classification performance to a more utility-based evaluation.

Keywords: Binary options, Utility-based evaluation, Kelly Criterion, Time series forecasting, Binary Classification, Calibration, Risk-adjusted returns.

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Jose Santos

*“The philosophers have only interpreted the world, in various ways.
The point, however, is to change it..”*

Karl Marx

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Abbreviations

API	Application Programming Interface
ATR	Average True Range (Volatility Indicator)
ART	Average ROI per Trade
AUC	Area Under the ROC (Receiver Operating Characteristic) Curve
DT	Decision Tree
EL	Expected Loss
EP	Expected Profit
EV	Expected Value
F1	Harmonic Mean of Precision and Recall
LDA	Linear Discriminant Analysis
LGBM	LightGBM
LR	Logistic Regression
MLP	Multi-Layer Perceptrons
NB	Gaussian Naive Bayes
QDA	Quadratic Discriminant Analysis
RF	Random Forest
ROC	Receiver Operating Characteristic
RSI	Relative Strength Index (Momentum Indicator)
SGD	Stochastic Gradient Descent
SMA	Simple Moving Average

Chapter 1

Introduction

Forecasting models are used widely across all areas to predict future states and guide decision-making, resource allocation and trade signals. When forecasting binary outcomes, the majority of models are compared on their ability to predict future values, using classification measures such as AUC, Accuracy, or F1 Score (Niculescu-Mizil and Caruana, 2005; Fawcett, 2006). These metrics provide significant information about the predictive ability of the model while not always representing the true real-world value of the predictions (Yardley and Petropoulos, 2021).

In various applications, studies have found that increasing forecasting accuracy often does not translate into greater financial utility or profitability (Yardley and Petropoulos, 2021). In part, this problem can be explained by the fact that there is often a gap between model predictions and the actual decisions made based on them. Because they do not take into consideration how forecasts are applied in practical activities, standard classification metrics do not represent the economic impact of predictions (Yardley and Petropoulos, 2021). This restriction is even worse in financial environments with fat-tailed payoffs, where results might be disproportionately impacted by rare but important events (Taleb, 2020). Even models with excellent classification accuracy could not be profitable in these situations.

The focus of this research is a case study in binary options trading. Binary options differ from typical financial instruments due to their binary nature and fixed payout structure (Cofnas, 2016). This unique characteristic offers an opportunity to investigate whether the observed disconnect between forecasting accuracy and utility in fat-tailed scenarios (Taleb, 2020) also applies to the binary and fixed nature of payoffs in binary options. To frame this problem, the forecasting task in this work will be approached as a binary classification problem, predicting upward or downward trends in the underlying asset's price.

To achieve its goals, this thesis incorporates state of the art methodologies in time series forecasting of binary events, utility-based evaluation and empirical analysis of trading strategies. By leveraging foundational principles from the literature, such as the Kelly Criterion for optimal bet sizing (Kelly, 1956) and risk-adjusted performance measures (Sharpe, 1966; Sortino and Price,

1994), the study evaluates the consistency and viability of different forecasting models for trading binary options.

1.1 Problem Statement

Our work contributes to the literature by studying the gap between forecasting classification accuracy and financial utility within the context of binary options trading. We frame the binary options trading task as a binary probabilistic classification problem. Particularly, we use supervised learning to predict upward or downward changes in asset prices based on a set of technical indicators (e.g., exponential moving average). We evaluate several classifiers using both classification and utility-based performance metrics, analyzing the relationship between the two types of measures. While there is a growing body of work highlighting the limitations of classification metrics in capturing real-world utility (Yardley and Petropoulos, 2021), the specific context of binary options has received limited attention.

While forecasting models are rigorously optimized for classification metrics e.g., accuracy, AUC, their practical utility, measured through risk-adjusted returns metrics or trading profitability, often remains unexamined. For instance, a model may achieve high AUC scores yet fail to generate profitable trading signals due to misaligned risk-reward dynamics or poor calibration of probability estimates (Niculescu-Mizil and Caruana, 2005). This discrepancy is particularly pronounced in domains with fat-tailed payoffs, where extreme events disproportionately affect outcomes (Taleb, 2020; Yardley and Petropoulos, 2021). However, it remains unclear whether this disconnect holds true in binary options trading, where payoffs are characterized by their binary nature and fixed payout (Cofnas, 2016).

This study aims to bridge the gap between forecasting accuracy and utility within the context of binary options trading. By framing the forecasting task as a binary classification problem, predicting upward or downward trends in asset prices, and evaluating whether classification metrics, align with trading performance in this setting, the research seeks to explore how the binary and fixed payout nature of these instruments influences the correlation between predictive accuracy and practical utility.

1.2 Research Questions

This study investigates the relationship between classification forecasting performance and financial utility in binary options trading through four core research questions:

- **RQ1: When classification metrics improves does utility also improve?** Does enhancing classification metrics reliably improve trading utility? We analyze models' predictive performance on the validation set and compare it with their practical utility measured during the test set backtesting period. Through pairwise model comparisons, we assess whether incremental gains in validation-time classification metrics consistently translate into better

test-time financial outcomes, revealing potential discontinuities between model optimization objectives and real-world trading performance 4.1.

- **RQ2: Do classification metrics align with financial performance?** How do classification metrics align with actual financial performance in binary options trading? We examine whether traditional metrics (e.g., AUC, accuracy) correlate with trading utility (e.g., risk-adjusted returns) under standard 75% payout conditions, testing the assumption that improved model accuracy necessarily translates to trading profitability 4.2.
- **RQ3: Do forecasting models maintain consistent rankings across classification and financial metrics?** To analyze if forecasting models maintain consistent rankings across metric families? We compare algorithm ranks on classification and utility based metrics 4.3.
- **RQ4: How does variation in binary options payout structures affect the relationship between accuracy and utility?** How sensitive are these relationships to varying payout structures? We replicate our analyses under extreme (60% and 90%) payout scenarios to determine whether the statistical-utility relationship persists across different market conditions 4.4.

1.3 Methodology and Contributions

This study investigates the misalignment between conventional classification metrics and actual financial performance in the domain of binary options trading. To examine this discrepancy, the forecasting problem is framed as a supervised binary classification task, predicting the short-term direction (up or down) of asset prices based on minute-level candlestick data and a diverse set of technical indicators.

The methodology begins with the transformation of financial time series data into a supervised learning classification problem. A dataset comprising Forex, commodities, cryptocurrencies, and stock prices was collected from the Polygon API, covering over 750,000 data points per asset. Technical indicators were computed for each data point, and machine learning models, including logistic regression, random forest, LightGBM, and neural networks, were trained and calibrated to generate probabilistic forecasts. These forecasts were then evaluated using both traditional classification metrics (e.g., AUC, F1 Score, Negative Log Loss) and utility-based metrics (e.g., Sharpe Ratio, Sortino Ratio, Average ROI per Trade). A backtesting framework (Subsection 3.2.6) was developed to simulate realistic trading conditions, containing payout structures, position sizing via the Kelly Criterion (Subsection 2.1.2), and real-world constraints typical of binary options platforms (Cofnas, 2016).

This thesis scientific contribution is its empirical investigation into the relationship between classification accuracy and financial performance in binary options trading. Through this study, it:

- Confirms insights from previous studies that classification accuracy does not necessarily lead to profitable trading decisions in binary options, (Niculescu-Mizil and Caruana, 2005; Yardley and Petropoulos, 2021; Taleb, 2020).
- Develops and applies a utility-aware evaluation framework, emphasizing metrics that reflect actual financial risk and return (Hand, 2012; Sokolova and Lapalme, 2009).
- It demonstrates the robustness of utility-based metrics across different market conditions and payout structures, challenging the conventional reliance on accuracy-focused model selection.
- It offers a reproducible pipeline for evaluating forecasting models in real-world speculative contexts, including calibration, backtesting, and risk-adjusted performance evaluation (Fawcett, 2006; Provost and Fawcett, 2013).

Together, these contributions highlight the importance of rethinking how forecasting models are evaluated and selected in real-world contexts where decisions have tangible consequences, models should be judged by their ability to drive better outcomes, not just their performance on classification metrics. The experiments and methods are fully reproducible and publicly available online.¹

1.4 Thesis Organization

Chapter 2 reviews the relevant literature on binary options, time series forecasting, classification and utility-based metrics. It establishes the theoretical foundation that guided this study.

Chapter 3 explains the methodology utilized in this study, detailing the data sources, feature engineering, and modeling framework. It also outlines how the forecasting task is transformed into a binary classification problem and describes the backtesting setup used to simulate trading scenarios.

Chapter 4 reports and analyzes the empirical results realized to answer the research questions 1.2. It compares models based on both statistical and utility-based metrics, highlighting the (mis)alignment between both metric groups.

Chapter 5 summarizes the thesis contributions, including the evaluation framework, methodological innovations, and key insights from the empirical analysis. It also outlines limitations and proposes future research directions for improving model evaluation in utility-focused financial forecasting.

¹<https://github.com/Jjoclas/forecast-utility-gap.git>

Chapter 2

Literature Review

By going over important ideas in binary options, time series forecasting and model evaluation, this chapter establishes the theoretical framework of this investigation. It starts with a summary of binary options trading, emphasizing its decision-making process, risk profile and mechanics. The discussion then moves to time series and forecasting binary events. Finally, the classification and utility-based metrics utilized in this work are explained, highlighting the distinction between predictive accuracy and real-world financial utility.

2.1 Binary Options

2.1.1 Definitions

Binary options are financial instruments that offer a fixed return based on a simple “yes” or “no” proposition: **Will the market price be above a certain level (the strike price) at the time of expiration?** (Cofnas, 2016)

Binary options trading has gained significant popularity, particularly since its approval in the US market in 2008 (U.S. Securities and Exchange Commission, 2008). Its appeal lies in its straightforward mechanics and high-profit potential (Securities and Commission, 2022). Unlike traditional trading methods, binary options require traders to predict only whether the market price of an asset will be above or below a predetermined strike price at the time of expiration.

Binary options are characterized by their binary nature, each trade results in one of two outcomes: success or failure. If the trader’s prediction is correct, they receive a predetermined payout, otherwise, they lose the amount invested. In some market implementations, a small fraction of the initial investment may be retained (e.g., 10%), but for the purposes of this study, we focus on the all-or-nothing scenario. (Securities and Commission, 2025; Cofnas, 2016)

The return (R) for a single binary option trade can be expressed as:

$$R = \begin{cases} \text{Payout} \times \text{Amount}, & \text{if the operation is successful} \\ -\text{Amount}, & \text{if the operation fails} \end{cases}$$

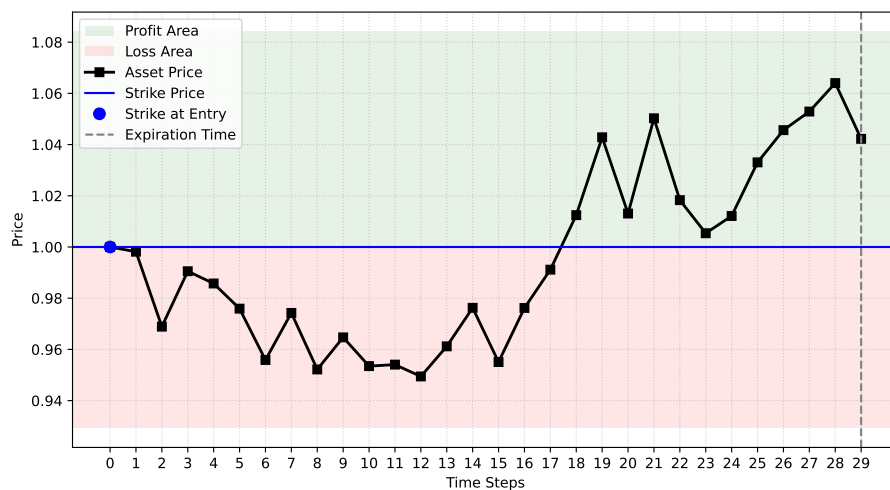


Figure 2.1: Illustration of a "call" binary option setup. The black line shows the asset's price over time. The blue line marks the strike price. The green and red shaded regions represent profit and loss areas, respectively. The vertical dashed line indicates the expiration time.

Where:

- **Payout:** The fixed multiplier is applied when the prediction is correct.
- **Amount:** The money invested in the operation.

Binary options are generally classified into two types: Call and Put options. A Call option (or Buy position) is used when the trader predicts that the market price will be above the strike price at the time of expiration. Conversely, a Put option (or Sell position) reflects the belief that the market price will be below the strike price at expiration (Cofnas, 2016).

The capital risked on a given binary trade is limited to the amount invested, but the payout is also predetermined, making binary options easy to understand but potentially risky due to their all-or-nothing nature.

Example Scenario: Imagine you invest \$100 in a binary option with a payout rate of 70%. If you predict correctly, you earn \$170 (initial investment + \$70 profit). If incorrect, you lose the \$100 invested.

2.1.2 Trading Binary Options

Bankroll Management

Bankroll management refers to the process of strategically allocating and sizing capital to balance potential returns with the risk of loss (Vince, 1992; ?). In trading that involves the strategic allocation of capital to maximize returns while minimizing the risk of significant losses (Vince, 1992). Effective bankroll management ensures that traders can sustain themselves through periods of volatility and uncertainty, allowing them to capitalize on profitable opportunities over the long

term. The key to successful trading lies not only in identifying profitable trades but also in managing the size of those trades relative to the total capital available (Wójcik, 2017). Poor bankroll management can lead to overexposure, emotional decision-making, and ultimately, the depletion of trading capital.

In the context of binary options trading, where outcomes are binary and risks are inherently high, bankroll management becomes even more crucial. Traders must carefully balance the size of their investments with the probability of success and the potential payout of each trade. Without a disciplined approach to managing their bankroll, traders risk losing their entire capital in a short period, even if their trading strategy is theoretically sound (Venter and De Jongh, 2022). To address this challenge, this paper adopts the Kelly Criterion, developed by J. L. Kelly Jr. in 1956. It provides a systematic method for determining the optimal fraction of a bankroll to allocate to each trade, balancing the trade-off between risk and reward to maximize long-term capital growth. (Kelly, 1956).

Kelly Criterion

In order to have a framework that enables a mathematical way to determine the optimal fraction of the bankroll to be placed on any given trade, this work utilizes the Kelly Criterion. This criterion is particularly relevant in scenarios when you have access to probabilistic information about the outcomes of events. The Kelly Criterion balances the trade-off between risk and reward by optimizing the expected logarithmic growth of wealth, avoiding ruin while maximizing returns (Kelly, 1956).

The Kelly Criterion is derived from the concept of maximizing the exponential rate of capital growth, denoted as G . For a binary outcome with two possible results (e.g., success or failure), the optimal fraction f^* of the bankroll to bet is given by:

$$f^* = \frac{bp - q}{b}$$

where:

- b is the net odds received on the bet (i.e., the payout ratio minus 1),
- p is the probability of winning,
- $q = 1 - p$ is the probability of losing.

This formula ensures that the trader bets a fraction of their bankroll proportional to the edge they have over the market, while avoiding overbetting.

In the context of binary options trading, the Kelly Criterion can be adapted to determine the optimal amount to invest in each trade based on the predicted probability of success and the payout ratio. For a binary option with a payout multiplier P_{payout} and a predicted probability of success P_{success} , the optimal fraction f^* of the bankroll to invest is:

$$f^* = \frac{P_{\text{success}} \cdot (\text{Payout} - 1) - (1 - P_{\text{success}})}{\text{Payout} - 1}$$

In this study, the raw Kelly output is further multiplied by an *aggressiveness factor* to manage risk exposure. This scalar adjusts how much of the theoretically optimal bet size is actually placed. We set this factor to 0.1, adopting a conservative approach that reduces the volatility and drawdown risk typically associated with full Kelly betting. Thus, the final fraction of capital invested in each trade is $f = 0.1 \cdot f^*$, which allows more stable capital growth over time.

In conclusion the Kelly Criterion provides a mathematically approach to bet sizing. By optimizing the expected logarithmic growth, it offers a systematic way to balance risk and reward in binary options trading. However, its effectiveness depends on accurate probability estimates.

2.2 Time Series

Time series analysis is an important financial market instrument, particularly for forecasting and trading strategy. A time series is a sequence of data points recorded over a specific interval of time (usually at equal time intervals). Financial market time series data typically have prices, volumes, and other indicators of markets that vary over time. From the analysis of these data, traders and researchers can identify patterns, trends, and potential future price directions of assets (Hyndman and Athanasopoulos, 2018).

2.2.1 Definitions

A time series is formally defined as a set of observations $\{y_t\}_{t=1}^T$, where y_t represents the value of the series at time t , and T is the total number of observations. Time series data can be univariate (a single variable observed over time) or multivariate (multiple variables observed simultaneously). In financial markets, common examples of time series include:

- **Price Series:** The historical prices of an asset, such as stock prices, currency exchange rates, or commodity prices.
- **Volume Series:** The trading volume of an asset over time, which reflects market activity.
- **Indicators:** Derived metrics such as moving averages, volatility measures, or technical indicators like the Relative Strength Index (RSI).

Time series analysis aims to model the underlying structure of the data to make predictions or infer relationships. This involves identifying components such as trends, seasonality, and noise, and using statistical or machine learning methods to capture these patterns (Hyndman and Athanasopoulos, 2018).

2.2.2 Forecasting Binary Events

Forecasting on time series data differs from other classification or regression tasks because it needs to account for temporal dependencies, trends, seasonality, and potential variations on the data distribution over time. Effective forecasting models aim not only to minimize prediction error but also to capture the dynamics that drive the temporal evolution of the variable of interest minimizing any existing noise (Dama and Sinoquet, 2021).

Forecasting binary events involves predicting the occurrence of one of two mutually exclusive outcomes. In financial markets, this typically refers to directional predictions, such as whether the price of an asset will go up or down over a specified time horizon. These types of forecasts are especially relevant in domains like binary options trading, where financial outcomes depend entirely on the accuracy of such directional signals (Cofnas, 2016).

Formally, the objective is to estimate the conditional probability $P(Y = 1 \mid \mathbf{x}_t)$, where $Y \in \{0, 1\}$ denotes the binary outcome (e.g., price increase or decrease) at time $t + 1$, and $\mathbf{x}_t$ represents the feature vector derived from historical market data up to time t . These features are inherently time-dependent, capturing temporal patterns such as trends, momentum, volatility, or seasonality. Models that forecast binary events learn to associate such temporal structures in $\mathbf{x}_t$ with the likelihood of directional movements in the immediate future (Hyndman and Athanasopoulos, 2018; Gu et al., 2020).

2.3 Calibration

An important aspect of forecasting binary events is not just assigning class labels (e.g., “up” or “down”), but providing well-calibrated probabilities that reflect the true likelihood of each outcome. Calibration refers to the agreement between predicted probabilities and observed frequencies. A perfectly calibrated model would ensure that among all predictions where the model assigns a probability of 0.7 to the positive class, the event actually occurs approximately 70% of the time (Guo et al., 2017).

Well-calibrated probabilities are particularly valuable in decision-making scenarios that depend on confidence levels, such as bet sizing in trading or threshold setting in risk management. Poorly calibrated models may output overconfident or underconfident predictions, which can mislead downstream applications even if their classification accuracy is high (Niculescu-Mizil and Caruana, 2005).

Calibration can be assessed and improved using techniques such as Platt Scaling, Isotonic Regression, or temperature scaling (Guo et al., 2017). These methods adjust the model’s raw output probabilities post-training to better align with empirical outcome frequencies, without altering the classifier’s decision logic.

In sum, forecasting binary events in financial time series involves not only learning accurate classification boundaries but also generating probability estimates that are both discriminative and

reliable. This dual requirement underscores the importance of combining predictive modeling with robust calibration strategies when dealing with binary financial decisions.

2.4 Evaluation Metrics

2.4.1 Classification Metrics

When evaluating the performance of predictive models in binary options trading, classification metrics are crucial. These metrics provide measures of a model's accuracy, reliability and ability to generalize to unseen data. The metrics chosen for this study are: Area Under the Curve (AUC), Accuracy, and F1 Score. These metrics are used in classification tasks and can provide insights into the quality of the model's forecasts (cf. [Fawcett \(2006\)](#); [Niculescu-Mizil and Caruana \(2005\)](#); [Sokolova and Lapalme \(2009\)](#); [Hand \(2012\)](#)).

Area Under the Curve (AUC)

The Area Under the ROC Curve (AUC) evaluates the model's ability to distinguish between positive and negative classes. This statistic comes from the Receiver Operating Characteristic (ROC) curve, which plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold settings. The AUC ranges from 0 to 1, where a value of 1 indicates perfect classification, and a value of 0.5 suggests no discriminative power (equivalent to random guessing) ([Fawcett, 2006](#)).

In the context of binary options trading, AUC is helpful for assess how reliably can the model distinguish between upward and downward price movements ([Hand, 2012](#)).

Accuracy

Accuracy is one of the most intuitive metrics for evaluating classification models. It measures the proportion of correct predictions (both true positives and true negatives) out of the total number of predictions. Mathematically, accuracy is defined as:

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{True Positives} + \text{True Negatives} + \text{False Positives} + \text{False Negatives}}$$

While accuracy is easy to interpret, it can be misleading in imbalanced datasets, where one class significantly outweighs the other. For example, in binary options trading, if the market has a strong tendency to move in one direction (e.g., upward), a model that always predicts "up" may achieve high accuracy but fail to capture meaningful patterns in the data. This is particularly problematic because such a model would be useless for identifying opportunities when the market moves in the opposite direction. Therefore, accuracy should be used in conjunction with other metrics, such as AUC and F1 Score, to provide a more comprehensive evaluation of model performance ([Provost and Fawcett, 2013](#)).

F1 Score

The F1 Score is a harmonic mean of precision and recall, which gives accurate representation of how well a model works. Precision measures the proportion of true positives among all predicted positives, while recall measures the proportion of true positives among all actual positives.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

The F1 Score is then defined as:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

The F1 Score is very helpful when there is an imbalance in the classes because it finds the right balance between precision and recall. The F1 Score can help you figuring out how well a binary options trading model can predict price movements in both directions, making sure that it doesn't favor one direction over the other (Sokolova and Lapalme, 2009).

Log Loss

The Log Loss (also called logarithmic loss or cross-entropy) is a performance metric that evaluates probabilistic predictions by penalizing lower confidence in correct classifications. Log Loss assesses the quality of predicted probabilities by penalizing incorrect classifications based on the confidence of the prediction. This makes it particularly useful for evaluating models in binary options trading, where probabilistic forecasts (e.g., the likelihood of an asset's price exceeding a strike price) are critical for decision-making (Fawcett, 2006).

For binary classification with true labels $y_i \in \{0, 1\}$ and predicted probabilities $\hat{p}_i$, the Log Loss is defined as:

$$L(y_i, \hat{p}_i) = -[y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)]$$

For a dataset with N observations, the average Log Loss is:

$$\text{Log Loss} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)]$$

In the context of binary options trading, Log Loss aligns with the goals of utility-based evaluation. It takes the prediction confidence in consideration, which is central to strategies like the Kelly Criterion that use predicted probabilities to size bets. Different from accuracy, which treats all errors equally, Log Loss rewards high confidence predictions when correct, and heavily penalized when wrong (Fawcett, 2006). Which makes it a good candidate for use as a training objective in Binary Options trading scenarios.

2.4.2 Utility-based Evaluation Metrics

While standard classification provide valuable insights into the performance of predictive models, they do not always reflect the real-world utility of a trading strategy. As highlighted by Taleb (2020), there can be significant differences between binary forecasts and actual payoffs in financial markets. Classification metrics often assume stable data distributions and fail to account for the dynamic and non-stationary nature of financial markets. For example, a model with high accuracy or AUC may still fail to deliver consistent profits in live trading due to factors such as transaction costs, market volatility, or changes in market conditions.

Utility-based evaluation metrics, in the context of binary options trading, measure the financial outcomes of trading decisions, providing a more realistic assessment of a strategy's profitability and risk by incorporating the actual payoffs and losses associated with trades. By combining classification and utility-based metrics, traders can, with better confidence, evaluate the model's performance and its potential effectiveness in real-world trading scenarios.

Expected Profit and Expected Loss (EPEL)

The Expected Profit (EP) and Expected Loss (EL), collectively referred to as EPEL, are utility-based metrics that separately quantify the potential gains and losses of a binary options trading strategy. These metrics are particularly useful for traders who need to evaluate the risk-return profile of their strategies in a more granular way (Venter and De Jongh, 2022).

The Expected Profit (EP) and Expected Loss (EL) for a binary options trade are defined as follows:

$$EP = P_{\text{success}} \cdot (\text{Payout} \cdot \text{Amount})$$

$$EL = P_{\text{failure}} \cdot \text{Amount}$$

Where:

- EP is the expected profit of the binary operation.
- EL is the expected loss of the binary operation.
- P_{success} is the probability of the operation being successful.
- $P_{\text{failure}} = 1 - P_{\text{success}}$ is the probability of the operation failing.
- **Payout** is the fixed multiplier for a successful trade.
- **Amount** is the capital invested in the operation.

The EPEL metrics allow traders to independently assess the potential profitability and risk of a trading strategy. By comparing the expected profit to the expected loss, traders can evaluate if the the reward of entering a operation outweighs the it's risk.

Expected Value (EV)

Building on the EPEL framework, the Expected Value (EV) is a metric that evaluates the overall potential outcome of a binary options trade, it is directly derived from the EPEL metrics, as it represents the difference between the Expected Profit (EP) and the Expected Loss (EL).

The Expected Value (EV) for a binary options trade is defined as:

$$EV = EP - EL$$

Substituting the formulas for EP and EL, we obtain:

$$EV = P_{\text{success}} \cdot (\text{Payout} \cdot \text{Amount}) - P_{\text{failure}} \cdot \text{Amount}$$

Where:

- EV is the expected value of the binary operation.
- P_{success} is the probability of the operation being successful.
- $P_{\text{failure}} = 1 - P_{\text{success}}$ is the probability that the operation fails.
- **Payout** is the fixed multiplier for a successful trade.
- **Amount** is the capital invested in the operation.

The Expected Value (EV) is a measure that evaluates the potential outcome of a binary options trade, considering the probabilities of success. It provides a mathematical framework for assessing the viability of using forecast values as a trading strategy under specific market conditions. Traders can use the EV to compare different trading strategies and select those with the highest expected profitability.

The effectiveness of utility-based evaluation metrics, such as Expected Profit and Expected Loss (EPEL) and Expected Value (EV), is highly dependent on the calibration (Section 2.3) of the predictive model. Poorly calibrated models can lead to misleading evaluations of trading strategies, as the expected outcomes may not reflect reality (*cf.* Brier (1950))

Sharpe Ratio

The Sharpe Ratio, introduced by William F. Sharpe in 1966, measures the excess return per unit of risk, where risk is defined as the standard deviation of returns (Sharpe, 1966). Mathematically, the Sharpe Ratio is defined as:

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

Where:

- R_p is the expected return of the portfolio or strategy,

- R_f is the risk-free rate (e.g., the return on government bonds),
- σ_p is the standard deviation of the portfolio's returns, representing total risk.

A higher Sharpe Ratio indicates better risk-adjusted performance, as it reflects higher returns relative to the volatility of the strategy. In the context of binary options trading, the Sharpe Ratio can help traders assess whether the returns generated by their strategy justify the level of risk taken (Sharpe, 1966). While the Sharpe Ratio assumes that returns are normally distributed and return of a given Binary Option is discrete and bimodal, the daily return resultant of the aggregation of all trades performed tend to approximate a normal distribution due to the Central Limit Theorem (?). This supports the use of the Sharpe Ratio to evaluate trades strategies under a Binary Options environment.

Sortino Ratio

The Sortino Ratio, a variation of the Sharpe Ratio, addresses some of its limitations by focusing only on downside risk. Unlike the Sharpe Ratio, which considers total volatility, the Sortino Ratio penalizes only negative deviations from the target return (Sortino and Price, 1994). It is defined as:

$$\text{Sortino Ratio} = \frac{R_p - R_t}{\sigma_d}$$

Where:

- R_p is the expected return of the portfolio or strategy,
- R_t is the target or minimum acceptable return (e.g., the risk-free rate),
- σ_d is the standard deviation of negative returns (downside deviation).

The Sortino Ratio is particularly useful for evaluating binary options trading strategies, as it focuses on the risk of losses rather than overall volatility. This aligns well with the binary nature of options trading, where the primary concern is avoiding significant drawdowns while maximizing returns. A higher Sortino Ratio indicates better performance relative to the risk of losses (Sortino and Price, 1994).

Both the Sharpe and Sortino Ratios provide valuable insights into the risk-adjusted performance of trading strategies. However, they serve slightly different purposes, the Sharpe Ratio is more general and suitable for evaluating strategies where both upside and downside volatility are relevant. The Sortino Ratio is more focused on downside risk, making it ideal for strategies where minimizing losses is a priority.

Average ROI per Trade (ART)

Traditional financial performance metrics, such as Return on Investment (ROI) and Cumulative Return, evaluate profitability relative to the initial amount of capital deployed. ROI is typically

calculated as the net gain divided by the initial investment amount, serving as a basic measure of efficiency (Ichsani and Suhardi, 2015). Similarly, Cumulative Return measures the total return over a period relative to the starting capital, without explicitly considering the capital dynamics of frequent trading (Investopedia, 2020).

However, in the context of binary options trading, traders continually reinvest capital across multiple short-duration trades. Traditional ROI and Cumulative Return fail to capture how efficiently models manage capital when it is risked multiple times during the trading period.

To better reflect the dynamics of frequent capital deployment, we adopt a more granular metric: *Average ROI per Trade (ART)*. This metric is adapted from the broader concept of ROI and was previously employed in trading studies such as (Ichsani and Suhardi, 2015). ART measures the average return per trade relative to the capital risked in that specific trade, providing a clearer picture of capital efficiency at the trade level.

Rather than expressing total profitability over time, ART captures the mean return relative to the risked capital on a per-trade:

$$ART = \frac{1}{n} \sum_{i=1}^n \left(\frac{\text{Net Return}_i}{\text{Capital Risked}_i} \right)$$

Where:

- Net Return_{*i*} is the profit or loss realized from trade *i*.
- Capital Risked_{*i*} is the capital amount invested in trade *i*.
- *n* is the total number of trades.

This formulation evaluates how well each unit of risked capital performs, offering a measure of trade efficiency, especially useful when comparing models with different position sizing behaviors.

While the Sortino and Sharpe ratios evaluate a model's economic value based on return volatility over a time window, they do not capture how efficiently a model manages its risk capital across trades. Two models may achieve similar profits, but one might do so with significantly lower capital exposure per trade.

In this way, ART serves as a valuable complementary utility metric, particularly suited for high-frequency and reinvestment-intensive domains such as binary and digital options trading.

2.5 Discrepancy between Classification Metrics and Utility

Binary classification methods are frequently applied to financial time series data for tasks such as predicting short-term directional price movements (Gu et al., 2020; Hyndman and Athanasopoulos, 2018). Models including logistic regression, decision trees, random forests, and neural networks are commonly used to estimate whether asset prices will move up or down over a specified horizon. These models are typically evaluated using statistical metrics such as Accuracy, AUC, and F1 Score, which offer a statistical approach to evaluate the model performance (Fawcett, 2006; Sokolova and Lapalme, 2009; Hand, 2012).

[Yardley and Petropoulos \(2021\)](#) defend that forecast evaluation must go beyond conventional error metrics and utilize methods to evaluate practical utility and economic impact. Their work emphasize that relying only on accuracy-based metrics can be misleading, especially when the real-world costs of false positives and false negatives are asymmetric or when operational decisions depend on forecast confidence. This perspective is most relevant in domains like finance, where incorrect predictions, even if statistically correct, can have disproportionate economic consequences.

Their study highlights several domains where utility-based evaluation frameworks have been applied, including finance, supply chain management, and energy forecasting. In finance, forecast accuracy is not always aligned with trading performance. In supply chain management, demand forecast accuracy does not always translate into optimal inventory control. One particularly relevant domain cited in the study is betting markets, where forecast accuracy has limited value unless it translates into profitable bets ([Yardley and Petropoulos, 2021](#)). Like binary options, betting markets are structured around fixed payouts and all-or-nothing outcomes, making the economic value of a prediction dependent not just on correctness, but also on the estimated probability and the implied odds.

A review of the literature reveals a significant gap: while utility-based evaluation frameworks have been explored in other domains, they remain underutilized in the specific context of binary options. Most studies in this area continue to focus on improving classification accuracy without assessing whether such improvements yield actual economic benefits. This lack of alignment between classification performance and utility-based outcomes represents an important and under-explored area for research.

The present work addresses this gap by explicitly evaluating both classification and utility-based metrics in the context of binary options trading. It highlights how different models, despite similar classification performance, can generate vastly different financial results, underscoring the importance of bridging the gap between model accuracy and practical utility.

Final Remarks

A strong basis for modeling and forecasting market direction using a variety of statistical and machine learning approaches has been established by the literature on binary classification in financial time series. Evaluation of these models has traditionally relied on classification metrics, which help quantify model performance in abstract terms. However, as highlighted by recent research, including ([Yardley and Petropoulos, 2021](#)), these metrics do not always align with the real-world utility of forecasts, particularly in decision-centric environments like trading.

In domains such as binary options, characterized by fixed payouts and all-or-nothing outcomes ([Option, 2025a](#)), there is a critical need to assess models based on their financial impact, not just their predictive correctness. Despite parallels with well-studied areas like betting markets, the literature still lacks comprehensive studies on the gap between classification evaluation and practical utility in binary options trading. This thesis aims to contribute to this underexplored

area by combining classification modeling with utility-based evaluation, offering a more decision-aware perspective for forecasting in speculative financial environments.

Chapter 3

Methodology and Materials

This chapter presents the data sources, feature construction process, modeling techniques, and evaluation framework used in this study. It outlines how financial time series data were processed, how the forecasting task was framed as a binary classification problem, and which machine learning models were employed. In addition, it introduces the backtesting setup developed to simulate binary option trading scenarios.

3.1 Dataset

The dataset used for this study was obtained from Polygon API, a reliable platform to obtain high resolution financial market data. It provides detailed candlestick data for a diverse range of intervals for various assets, which is critical for time series analysis and building robust forecasting models ([Polygon.io, 2025](#)).

3.1.1 Financial Assets

The dataset includes minute-by-minute candlestick data for various assets, categorized into Forex currency pairs, commodities, cryptocurrencies, and indices.

Forex Currency Pairs

Symbol	Description
EURUSD	Euro to US Dollar
USDJPY	US Dollar to Japanese Yen
GBPUSD	British Pound to US Dollar
AUDUSD	Australian Dollar to US Dollar
USDCAD	US Dollar to Canadian Dollar
EURGBP	Euro to British Pound
EURJPY	Euro to Japanese Yen
GBPJPY	British Pound to Japanese Yen
AUDJPY	Australian Dollar to Japanese Yen
CHFJPY	Swiss Franc to Japanese Yen
EURCAD	Euro to Canadian Dollar
GBPCAD	British Pound to Canadian Dollar
NZDJPY	New Zealand Dollar to Japanese Yen
EURCHF	Euro to Swiss Franc
USDSGD	US Dollar to Singapore Dollar

Table 3.1: List of Forex currency pairs included in the dataset

Commodities

Symbol	Description
XAUUSD	Gold to US Dollar
XAGUSD	Silver to US Dollar

Table 3.2: List of commodities included in the dataset

Cryptocurrencies

Symbol	Description
BTCUSD	Bitcoin to US Dollar
ETHUSD	Ethereum to US Dollar
LTCUSD	Litecoin to US Dollar
XRPUSD	Ripple to US Dollar

Table 3.3: List of cryptocurrencies included in the dataset

Stocks

Symbol	Description
AAPL	Apple Inc.
MSFT	Microsoft Corporation
AMZN	Amazon.com Inc.
GOOGL	Alphabet Inc. (Google)

Table 3.4: List of stocks included in the dataset

The data span from 29 March 2023 to 17 March 2025, capturing more than 750K data points per asset.

3.1.2 Data Structure

Each record in the dataset corresponds to a single candlestick, representing price movements and trading activity within a one-minute interval. The detailed structure of the data set, including the description of features and data types, is summarized in Table 3.5.

Table 3.5: Features and descriptions of the candlestick dataset.

Feature Name	Data Type	Description
from	INTEGER	The UNIX timestamp (in seconds) marking the beginning of the candlestick. <i>Example: 1622203200 (specific date/time).</i>
at	INTEGER	The UNIX timestamp (in nanoseconds) of the latest update to the candlestick. <i>Example: 1622203200500000000.</i>
to	INTEGER	The UNIX timestamp (in seconds) marking the end of the candlestick. <i>Example: 1622203260.</i>
open	FLOAT	The value of the asset at the start of the candlestick period. <i>Example: 1.2200.</i>
close	FLOAT	The value of the asset at the end of the candlestick period. <i>Example: 1.2210.</i>
min	FLOAT	The lowest price of the asset during the candlestick period. <i>Example: 1.2195.</i>
max	FLOAT	The highest price of the asset during the candlestick period. <i>Example: 1.2215.</i>
volume	INTEGER	The total number of trades. <i>Example: 500 trades.</i>

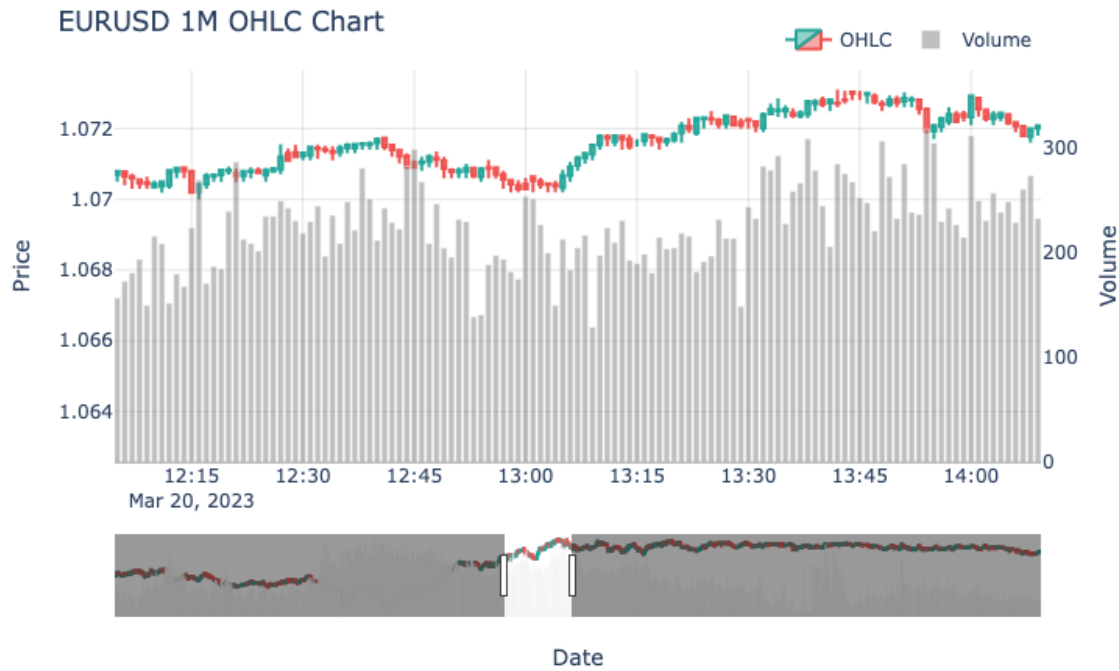


Figure 3.1: One-minute OHLC (Open-High-Low-Close) chart of EUR/USD with volume. The color indicates the price movement, green for a price increase (close > open) and red for a decrease (close < open). Volume its shown as verticals bars.

3.2 Methodology

The methodology for this study is designed to bridge the gap between forecasting accuracy and utility, focusing on the practical application of predictive models in binary options trading. The workflow integrates robust data preparation, advanced modeling techniques, and comprehensive evaluation metrics to optimize performance and ensure real-world applicability. The process is structured as follows:

3.2.1 Transforming Time Series Forecasting into a Binary Classification Problem

In this work, short-term price forecasting for binary options is treated as a binary classification problem. Rather than solving for a regression, the model is trained to predict the direction (*trend*) of the next candlestick: whether the close price will be higher or lower than it's open. This aligns directly with the payoff structure of binary options, which depends only on the asset moving above or below the strike price (Cofnas, 2016). By focusing on the sign of the price movement (up/down) instead of its magnitude, we tailor the problem to the decision context. For example, a highly accurate regression forecast that slightly overshoots may still lose money in binary options if it predicts the wrong direction, whereas a classification model explicitly trained on direction is directly rewarded for correct up/down guesses.

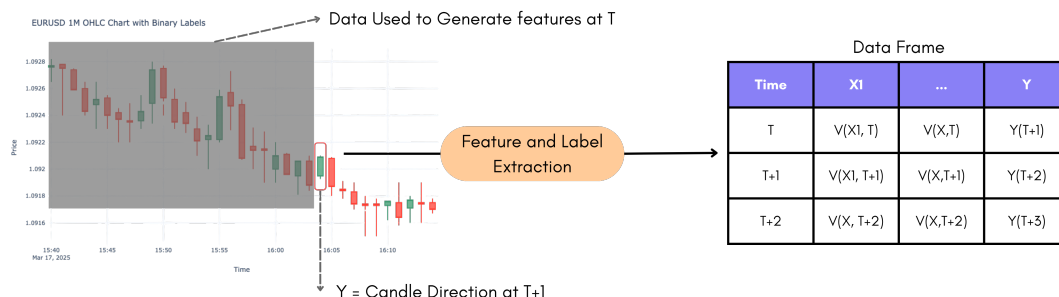


Figure 3.2: Pipeline for transforming raw time-series data into a supervised learning dataset, with features derived from technical indicators and binary labels indicating directional movement.

Each candlestick (at time t) yields a feature vector of technical indicators computed from recent price history (e.g., moving averages, momentum oscillators, etc.) and a label determined by the next candle: **Up (1)** if the close at $t+1$ is greater than the open at t , and **Down (0)** otherwise, whenever there is no direction (open and close price are equal), those data points are removed from the dataset, *This does not impact the backtesting results, as in platforms like IQ Option, if the closing price equals the strike price at expiration (i.e., the option expires at-the-money), the initial investment is returned to the trader (Center, 2025).* As illustrated in Figure 3.2, this process transforms a sequential time-series into a tabular dataset suitable for classification. The overall label distribution is can be visualized in Figure 3.3, where neutral (NaN) excluded values accounts for around 6.9% of the data.

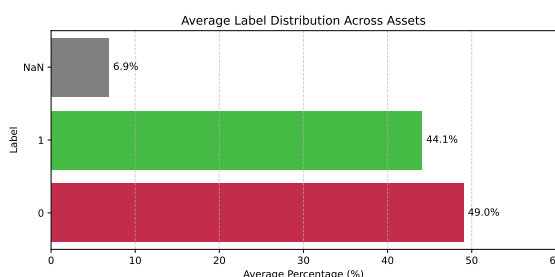


Figure 3.3: Average label distribution across all assets.

From a modeling perspective, this approach invokes standard classification techniques. We compute a set of technical indicators from the candlestick series (such as moving averages, RSI, MACD, volatility measures, etc.) under the assumption that these derived features contain predictive patterns in the price time series (Hyndman and Athanasopoulos, 2018). A classification model (e.g., logistic regression, random forest, neural network) can then learn a decision boundary in the indicator space separating up-moves from down-moves. In other words, instead of fitting a continuous forecast function $y_{t+1} = f(y_t, y_{t-1}, \dots)$, we learn $P(\text{Up}|\mathbf{x}_t)$ for indicator vector $\mathbf{x}_t$, and threshold it for a directional decision.

This reframing offers two key advantages. First, it simplifies the target to the essential outcome for binary options. Only the direction of movement matters to profitability, not the magnitude.

Second, it allows the use of well-established classification evaluation metrics and calibration techniques. Metrics such as accuracy, F1 score, and AUC evaluate how well the model distinguishes up vs. down moves (Hand, 2012; Fawcett, 2006). Furthermore, probabilistic calibration ensures that the model's estimated probabilities align with observed success frequencies, a critical feature when translating predictions into trade sizing and expected value (Niculescu-Mizil and Caruana, 2005).

Key steps in the transformation pipeline include: computing indicator features from each historical candle, assigning a binary label based on the subsequent candle's outcome, and assembling these into a classification dataset. Figure 3.2 summarizes this pipeline. This pipeline is based on the assumption that temporal patterns and technical signals in recent price history carry information about the next period's direction. While markets are noisy, casting the problem as classification focuses the model on robust directional signals, if they exist. This framing is justified by the nature of binary options: given their all-or-nothing payoff, predicting the sign of price change becomes the primary modeling objective.

3.2.2 Data Extraction and Preparation

Data extraction and preparation involve all necessary steps, from data collection, normalization, feature engineering, till the transformation of the original time series forecasting task into a binary classification problem, all steps were realized as it follows:

- Data Collection: Fetch minute-by-minute candlestick data for the selected forex pairs from the Polygon API.
- Data Splitting: The dataset is divided into three distinct periods, check Figure 3.4:
 - Training Set: Data prior to June 1st, 2024
 - Validation Set: Data from June 1st, 2024 to October 1st, 2024
 - Test Set: Data from October 1st, 2024 onwards
- Feature Engineering: Extract technical indicators and features that capture market behavior. A comprehensive list of technical indicators used in this study, along with their descriptions, is provided in Table 3.6.
- Normalization and Scaling: Standardize the data to ensure compatibility with machine learning models.

Table 3.6: Technical Indicators Used in the Study

Category	Indicator Name	Description
Date Features	HourOfDay	Hour of the day extracted from timestamp.
	DayOfWeek	Day of the week extracted from timestamp.
Overlapping Studies	SMA (5, 10)	Simple Moving Average over 5 and 10 periods.
	Bollinger Bands (Upper, Middle, Lower)	Band indicators based on 14-period SMA.
	EMA (5, 10, 50, 100, 200)	Exponential Moving Average over different periods.
	HT Trendline	Hilbert Transform Trendline.
Momentum Indicators	MACD	Moving Average Convergence Divergence.
	RSI	Relative Strength Index (14-period).
	Aroon Oscillator	Measures trend strength (10-period).
	ROC	Rate of Change (10-period).
Volatility Indicators	ATR	Average True Range (10-period).
	NATR	Normalized ATR (10-period).
	TRANGE	True Range indicator.
Volume Indicators	OBV	On-Balance Volume.
	AD	Accumulation/Distribution Line.
	MFI	Money Flow Index.
Pattern Recognition	CDL3Inside	Three Inside Up/Down candlestick pattern.
	CDL3Outside	Three Outside Up/Down candlestick pattern.
	CDLAdvanceBlock	Advance Block candlestick pattern.
	CDLBeltHold	Belt-hold candlestick pattern.
	CDLCloseMarubozu	Closing Marubozu pattern.
	CDLDoji	Doji candlestick pattern.
	CDLDojiStar	Doji Star pattern.
	CDLDragonflyDoji	Dragonfly Doji pattern.
	CDLEngulfing	Bullish/Bearish Engulfing pattern.
	CDLEveningStar	Evening Star candlestick pattern.
	CDLGravestoneDoji	Gravestone Doji pattern.
	CDLHammer	Hammer candlestick pattern.
	CDLHangingMan	Hanging Man pattern.
	CDLHaram	Harami candlestick pattern.
	CDLHaramCross	Harami Cross pattern.
	CDLHighWave	High-Wave candlestick pattern.
	CDLHikkake	Hikkake candlestick pattern.
	CDLIdentical3Crows	Identical Three Crows pattern.
	CDLInvertedHammer	Inverted Hammer pattern.
	CDLLongLeggedDoji	Long-Legged Doji pattern.
	CDLLongLine	Long Line candlestick pattern.
	CDLMarubozu	Marubozu candlestick pattern.
	CDLMatchingLow	Matching Low candlestick pattern.
	CDLRickshawMan	Rickshaw Man candlestick pattern.
	CDLSeparatingLines	Separating Lines pattern.
	CDLShootingStar	Shooting Star pattern.
	CDLShortLine	Short Line candlestick pattern.
	CDLSpinningTop	Spinning Top candlestick pattern.
	CDLStalledPattern	Stalled candlestick pattern.
	CDLTakuri	Takuri candlestick pattern.
	CDLThrusting	Thrusting candlestick pattern.
	CDLXSideGap3Methods	X Side Gap 3 Methods pattern.
Price Transforms	TSF	Time Series Forecast (10-period).
	VAR	Variance (10-period).

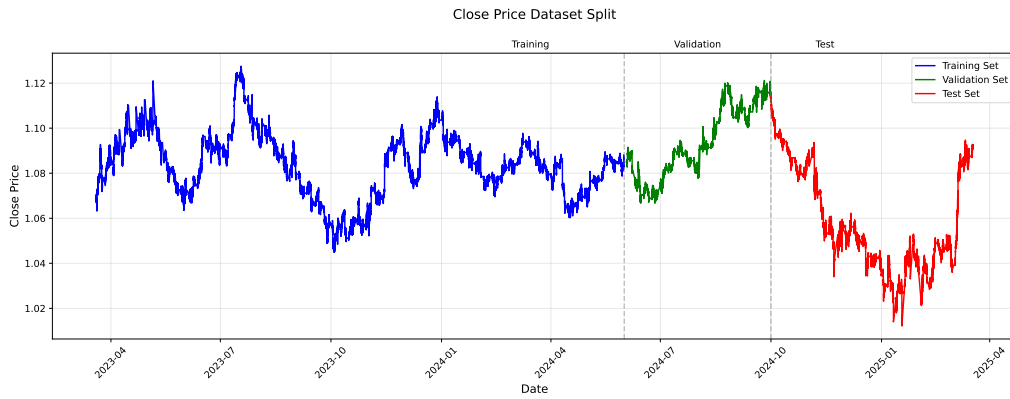


Figure 3.4: Chronological split of EUR/USD 1-minute close price data (March 2023 - March 2025) showing training, validation, and test periods. The vertical dashed lines demarcate the splits at June 1st 2024 (validation start) and October 1st 2024 (test start).

3.2.3 Model Training

The goal of model training is to learn predictive functions that estimate the probability of price movements. To align with this objective, the forecasting task is framed as a binary classification problem. Each time point is labeled as an upward movement (1) or a downward movement (0).

All models are trained using the negative log-loss(Subsection 2.4.1) as the objective function. To optimize predictive performance and reduce overfitting, a hyperparameter search is performed using grid search and cross-validation. This tuning process is executed independently for each model and for each asset, ensuring that the parameter configuration is tailored to the unique distributional characteristics of every financial instrument.

Finally, calibrated probability estimates are produced for all classifiers. Calibration techniques are applied to ensure that predicted probabilities reflect true likelihoods, a necessary condition for models to be useful in utility-based evaluation frameworks such as expected value and Kelly criterion sizing (Niculescu-Mizil and Caruana, 2005).

3.2.4 Learning Algorithms

We employ several classification algorithms including:

- **AdaBoost:** A boosting method technique that combines weak learners to improve predictive performance (e.g., Beja-Battais (2023); Singh (2022)).
- **Decision Tree (DT):** A simple tree-based model that splits the data features on rule based thresholds (e.g., Rokach and Maimon (2005); Singh (2022)).
- **Dummy Classifier (Dummy):** A baseline model that makes predictions using simple rules, for this study it predicts the most frequent class observed, is used for comparison purposes (e.g., scikit-learn (2025)).

- **Gaussian Naive Bayes (NB)**: A probabilistic classifier that assumes that features follow a normal distribution and makes predictions based on Bayes' theorem (e.g., Rish (2001); Tripathy and Behera (2023)).
- **Linear Discriminant Analysis (LDA)**: A classification technique that models that finds a linear combination of features to best separate the target classes (e.g., Tharwat et al. (2017)).
- **LightGBM**: A gradient boosting framework that uses histogram-based algorithms for fast and efficient tree learning (e.g., Ke et al. (2017)).
- **Logistic Regression (LR)**: A common linear model for binary classification, providing interpretable and calibrated probability outputs (e.g., Cox (1958); Zheng et al. (2025)).
- **Multi-Layer Perceptron (MLP)**: An artificial neural network capable of modeling complex non-linear relationships (e.g., Popescu et al. (2009); Zheng et al. (2025)).
- **Quadratic Discriminant Analysis (QDA)**: An extension of LDA that allows each class to have its own covariance matrix, enabling quadratic decision boundaries (e.g., Wu and Hao (2021)).
- **Random Forest (RF)**: An ensemble of decision trees trained on subsets of data and features to reduce overfitting and increase generalization (e.g., Breiman (2001); Singh (2022)).
- **Ridge Classifier (Ridge)**: A regularized version of linear regression for classification, applying L2 penalty to reduce variance and mitigate multicollinearity (e.g., Hoerl and Kennard (1970); Tripathy and Behera (2023)).
- **Stochastic Gradient Descent (SGD)**: An optimized linear classifier that scales well to large datasets using incremental updates (e.g., Nakkiran et al. (2019); Tripathy and Behera (2023)).

We use the implementations available on *scikit-learn* Python package (?), except for LightGBM for which we use the *lightgbm* Python package (Ke et al., 2017).

3.2.5 Evaluation Metrics

To evaluate the performance of predictive models, we implemented a backtesting framework that replicates the mechanics of short-term binary options trading on regulated and retail platforms such as IQ Option or Nadex. The simulation was designed to mirror the operational environment in which these models would be deployed, including strike mechanics, expiration conventions, capital management, and payout structures.

To ensure a comprehensive evaluation, this study employs a dual-metric framework encompassing both classification and utility-based measures. These metrics provide complementary perspectives on model effectiveness, one from a predictive perspective and the other from a financial decision-making perspective. Importantly, the same set of metrics will be utilized in Chapter 4.

- **Classification Metrics:**

- **Accuracy:** Proportion of correct predictions (both upward and downward price movements).
- **F1 Score:** Harmonic mean of precision and recall, particularly useful for imbalanced class distributions.
- **Area Under the ROC Curve (AUC):** Measures the model's discriminative ability across classification thresholds.
- **Negative Log Loss (NLL):** Evaluates the quality of predicted probabilities by penalizing incorrect high-confidence predictions. Higher values indicate better performance due to the negation.

- **Utility-Based Metrics:**

- **Backtesting Final Balance:** Is the return from the simulated cumulative profits and losses over the backtesting (Subsection 3.2.6) period, based on actual model predictions and a fixed payout structure.
- **Sharpe Ratio:** Quantifies risk-adjusted return by dividing the excess return by the standard deviation of returns.
- **Sortino Ratio:** A variation of the Sharpe Ratio that only penalizes downside volatility, aligning more closely with risk in binary options.
- **Average ROI per Trade (ART):** Measures profitability relative to the total amount of capital actually risked across all trades, accounting for dynamic position sizing.

3.2.6 Backtesting

To evaluate the practical performance of predictive models, a backtesting framework was implemented that emulates the mechanics of regulated digital/binary options platforms (e.g., IQ Option, Nadex) (Option, 2025b). The simulation incorporates key operational features documented in broker guides and regulatory materials, including strike price conventions, payout structures, and expiration mechanics.

3.2.6.1 Overall Steps

1. **Data Splitting:** Backtesting is conducted exclusively on the held-out test set, beginning on October 1st, 2024. As detailed in Subsection 3.2.2. This temporal split ensures that backtest results reflect out-of-sample performance, simulating real-time deployment conditions. Figure 3.4 illustrates the segmentation applied to the EUR/USD 1-minute close price series.
2. **Signal Generation & Trade Eligibility:** For each candlestick interval (1 minute), the model predicts price direction (call/put), if the next candlestick `close` price is going to be higher or lower than the its candle `open` price. A trade is executed only if the calculated expected value (EV) (Subsection 2.4.2) is positive, filtering low-confidence predictions.

3. **Strike Price Determination:** The strike is set to the `open` price of the next candle, mirroring brokers standard practice of using the spot price at trade initiation as the at-the-money strike (Securities and Commission, 2025).¹
4. **Expiration & Settlement:** The option expires at the end of the subsequent candlestick, with settlement determined by its `close` price. This simulates short-duration contracts (e.g., 1-minute to 30-minute expirations common on IQ Option (Option, 2025c)), where the outcome depends on the next immediate price movement. The simulation assumes instantaneous execution at candle `open`, though real-world latency between signal generation and order placement may affect results (Check Limitations 3.2.6.2).
5. **Payout Structure:** A fixed 75% return on successful trades is assumed, reflecting conservative payout rates for at-the-money options across major brokers (TradingPedia, 2023). While platforms may offer variable payouts (e.g., up to 900% for deep out-of-the-money strikes (Option, 2025c)). Losses are capped at 100% of the amount invested, per the all-or-nothing payoff structure (NASAA, 2017).
6. **Risk-Adjusted Position Sizing:** The Kelly Criterion dynamically adjusts the fraction of capital wagered per trade, optimizing long-term growth given the binary outcome probabilities and asymmetric payoff (75% gain vs. 100% loss). This addresses the "fixed risk, fixed reward" nature emphasized by regulators (CFTC and SEC, 2018), where optimal bet sizing is critical to managing drawdowns.
7. **Profit Calculation & Metrics:** Returns are aggregated across all trades, with Sharpe/-Sortino ratios computed to assess risk-adjusted performance. The simulation excludes slippage, focusing purely on model predictive accuracy under idealized execution (Check Limitations 3.2.6.2).

3.2.6.2 Key Limitations vs. Real-World Conditions:

The simulation framework used in this study simplifies certain aspects of real-world binary options trading. First, the use of candlestick `open` prices to determine the strike price, it assumes perfect temporal alignment between signal generation and execution. In practice, trading platforms introduce micro-latencies and slippage, specially for high-frequency “turbo” options where execution timing is critical (Option, 2025a). Second, the model employs a fixed 75% payout. While this simplification is representative of typical at-the-money contracts, actual broker platforms often vary payouts based on strike proximity and market volatility (Option, 2025c). Third, the simulated trades are held to expiration without the option of early closure. Some platforms permit early exit to recover part of the stake if market conditions reverse, a feature that could modestly reduce losses in live trading environments (Trade, 2018).

¹This replicates the "auto nearest strike" functionality described by IQ Option (Option, 2025b), where the platform automatically selects the closest available threshold to the current price.

Despite these limitations, the design choices reflect the structural norms of short-term digital options. Setting the strike price to the label candle's open and assuming expiration at the following interval captures the core mechanics of fast-paced, high-frequency retail trading platforms (CFTC and SEC, 2018). The fixed 75% payout corresponds to modest value of broker contracts (TradingPedia, 2023), and the use of Kelly criterion sizing introduces offers a mathematically, utility-driven method for capital allocation under binary risk, grounded in both gambling theory and financial optimization literature (Kelly., 1956).

Final Remarks

In summary, this chapter established the methodological framework for the study. The integration of calibrated probabilistic models, technical indicators, and a backtesting setup ensures that the performance assessments are both empirically grounded and practically meaningful. This framework supports a deeper investigation into the relationship between prediction accuracy and economic outcomes.

Chapter 4

Results

We conducted experiments to investigate the relationship between classification accuracy and financial utility in binary options trading. Our analysis addresses the four key research questions put forth in Section 1.2. For each research question, we analyze the performance of our models across both traditional classification metrics (Accuracy, AUC, F1 Score, Log Loss) and utility-based metrics (Final Balance, Sharpe Ratio, Sortino Ratio, ART). This approach enables us to systematically examine the relationship between classification and financial performance.

4.1 When classification metrics improves does utility also improve?

In the first research question, we analyse the degree of association between changes in classification performance and changes in utility. We compute pairwise differences in model rankings across metrics. For each ordered model pair in the set of models, we assess how their relative rank changes in both a classification metric (accuracy and negative log loss¹) and a utility metric (Backtesting final balance). A lower rank number indicates stronger performance; thus, if a model ranks second in accuracy compared to another ranked fifth, the first model performs better than the second. The analysis then examines whether this same model also achieves a better rank in when evaluated with an utility-based metric. Each comparison is visualized as a point in a two-dimensional space: the x-axis represents the change in rank for two classification metrics (accuracy, negative log loss), and the y-axis for the utility metric. This visualization allows us to assess whether gains in classification metrics align with corresponding improvements in financial utility. In summary:

- If a model has a lower rank number in a classification metric (e.g., Accuracy rank 2 vs 5), we interpret this as an improvement in classification performance.
- We then ask: does that same model also achieve a better rank (lower value) in a utility metric such as *Backtesting Final Balance*?

¹We use the negative of the log loss so that improvements (better performance) are represented by higher values, consistent with the utility metrics used.

Each pairwise comparison produces a point in a two-dimensional plot:

- The x-axis reflects how much better or worse a model is in the **classification metric**.
- The y-axis reflects how much better or worse it is in the **utility metric**.
- Upper-right: both classification and financial performance worsened.
- Lower-right: classification performance improved, but financial performance worsened.

This enables a clear visual test: do improvements in classification metrics coincide with better financial outcomes? Figure 4.1 and Figure 4.2 display two such scatter plots.

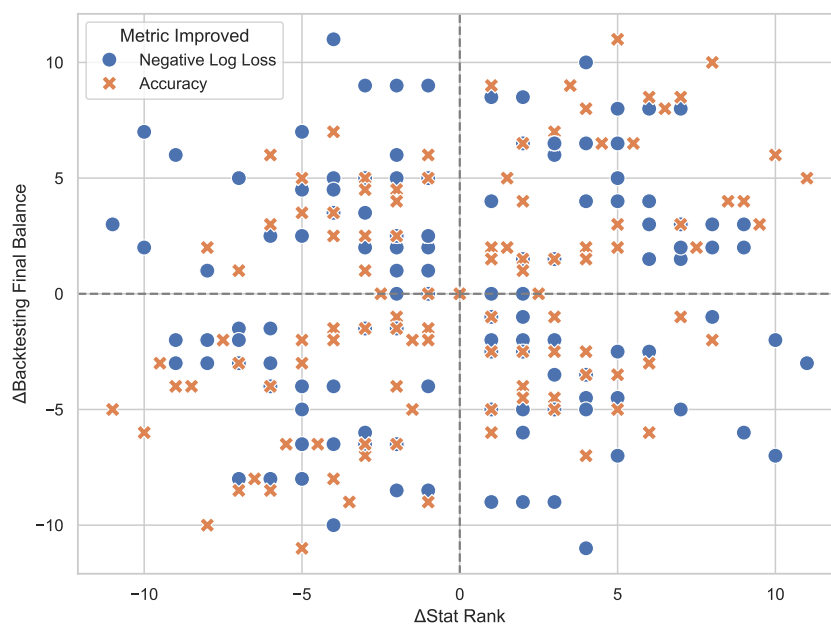


Figure 4.1: Change in Accuracy and Neg Log Loss vs. change in Backtesting Final Balance.

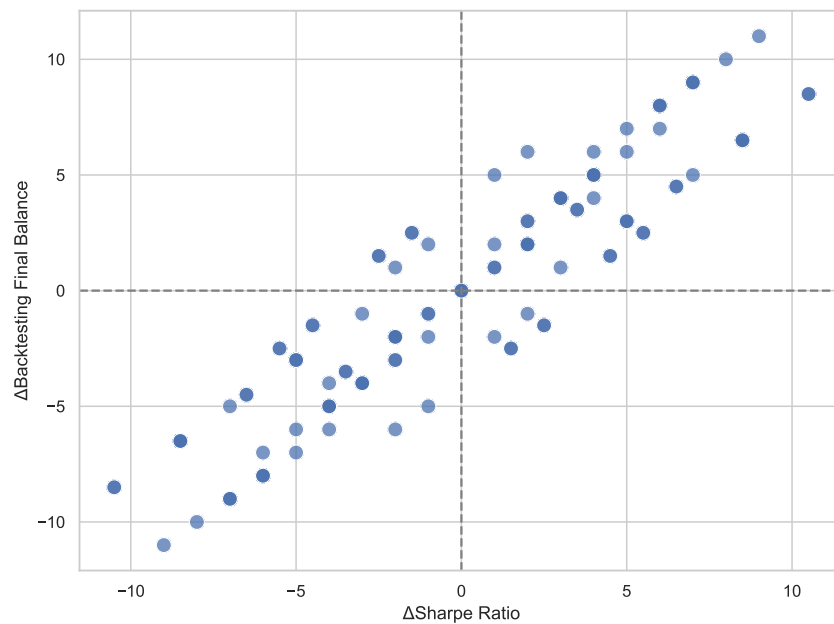


Figure 4.2: Change in Sharpe Ratio vs. change in Backtesting Final Balance.

The scatter plot 4.1 comparing Accuracy and Negative Log Loss improvements to changes in Backtesting Final Balance. The points are dispersed, showing no clear pattern. This result suggests that improvements in classification metrics do not necessarily correlate with better financial outcomes. In many cases a improvement in classification accuracy translated to worse financial performance.

In contrast, the scatter plot 4.2 comparing **Sharpe Ratio** to **Backtesting Final Balance** shows a strong and linear alignment: nearly all points lie along a 45-degree line from bottom-left to top-right. This pattern indicates that when a model's Sharpe Ratio improves, its Backtesting Final Balance usually improves too, conversely when the Sharpe Ratio worsens, so does its financial outcome.

These findings have important implications for model selection. The results suggests that classification performance metrics are insufficient predictors of financial success. Instead, utility-based metrics such as the Sharpe Ratio offer more reliable assessment of a model's performance in trading environments.

4.2 Do Classification Metrics Align with Financial Performance?

While RQ1 focused on analyzing pairwise differences between models to understand if improvements in classification metrics led to corresponding improvements in utility, RQ2 takes a broader view by examining the overall correlation between classification metrics (in the validation set) and financial performance (in the test set) across all models and assets. This allows us to determine whether classification metrics can be used as a proxy for real-world trading performance.

To analyze the relationship between validation and test metrics, we conduct two complementary correlation analyses, under a 75% payout structure. The first uses Pearson correlation coefficients to measure linear relationships between the values of validation and test metrics across all asset-model pairs. The second employs Spearman correlation coefficients to capture potentially non-linear but monotonic relationships between these metrics.

The results of this analysis are shown in two correlation matrices, shown in Figure 4.3 and Figure 4.4, which help us quantify the degree to which validation metrics are associated with real-world trading performance.



Figure 4.3: Pearson correlation matrix comparing validation metrics with test-time financial performance.

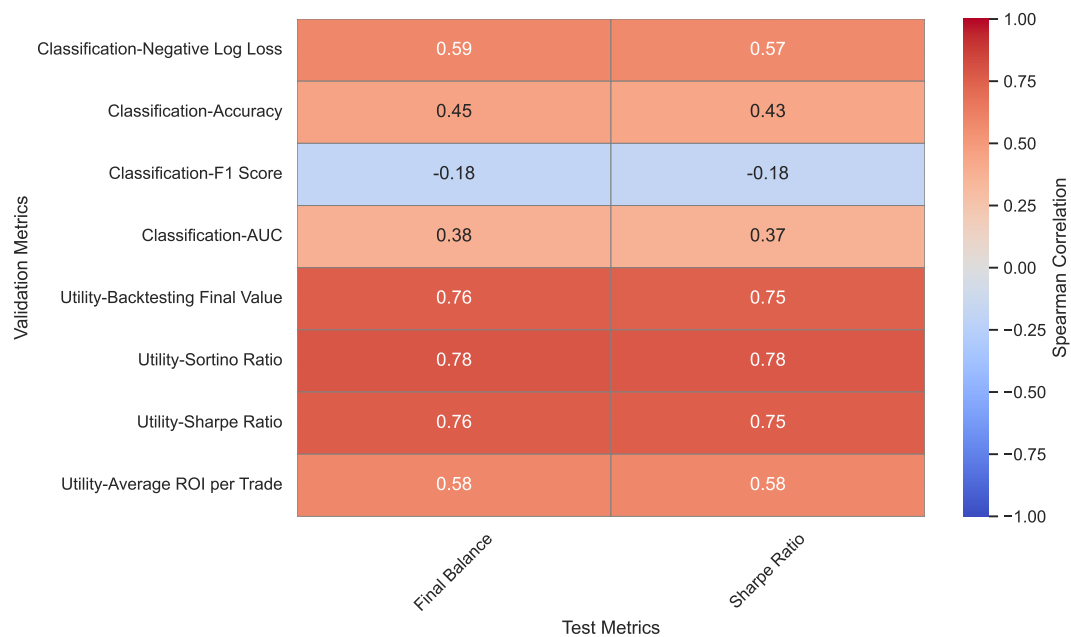


Figure 4.4: Spearman correlation matrix comparing validation metrics with test-time financial performance.

Our correlation analysis indicate that utility-based validation metrics, including Account Final Balance, Sharpe ratio, and Sortino ratio, show a stronger correlation with actual financial performance during backtesting relative to classification metrics.

Some metrics, such as F1 score show a correlation of -0.18, which denotes negative associations with financial outcomes. While Negative Log Loss performs better than other classification metrics, reaching correlations of 0.59 (Spearman) and 0.32 (Pearson), it is still considerably lower compared to the correlation of utility-based metrics.

These results have important implications for model selection in financial applications. Relying on classification metrics may lead to suboptimal choices and unreliable performance estimates. The results suggest that financially-oriented utility metrics provide more reliable indicators of a model’s potential performance in real-world trading scenarios. This underscores the importance of using domain-specific evaluation when developing models for financial applications.

4.3 Do Forecasting Models Maintain Consistent Rankings Across Classification and Financial Metrics?

To answer the third research question we analyzed whether models that perform well on traditional classification metrics also excel in real-world financial terms. In other words: *are the best models according to classification metrics also the most profitable to deploy in practice?*

Average Test-Set Rank per Model

Table 4.1 reports the average test-set ranking of each model for the four classification metrics. Lower ranks indicate better relative performance.

Table 4.1: Average rank and position (in parentheses) of each model across classification metrics.

Classifier	Log Loss	Accuracy	F1	AUC
AdaBoost	6.84 (8)	6.48 (6)	5.60 (2)	4.92 (4)
DT	5.52 (3)	6.66 (7)	7.60 (11)	6.36 (7)
Dummy	8.64 (11)	9.48 (12)	6.50 (7)	10.64 (12)
NB	7.44 (10)	8.44 (10)	6.76 (8)	8.08 (10)
LDA	5.48 (2)	4.72 (3)	6.22 (6)	4.88 (3)
LightGBM	7.16 (9)	8.22 (9)	7.50 (10)	7.00 (9)
LR	4.88 (1)	3.86 (1)	6.20 (5)	4.08 (1)
MLP	5.72 (5)	5.32 (5)	4.48 (1)	5.72 (5)
QDA	8.84 (12)	9.06 (11)	6.14 (4)	9.08 (11)
RF	6.08 (7)	6.66 (7)	7.82 (12)	6.56 (8)
Ridge	5.56 (4)	4.34 (2)	6.10 (3)	4.76 (2)
SGD	5.84 (6)	4.76 (4)	7.08 (9)	5.92 (6)

Table 4.2: Average rank and position (in parentheses) of each model across financial utility metrics.

Model	Final Balance	Sortino Ratio	Sharpe Ratio	ART
AdaBoost	7.14 (11)	6.98 (10)	7.10 (11)	6.94 (11)
DT	5.88 (2)	5.56 (2)	6.08 (3)	6.68 (9)
Dummy	6.46 (5)	6.24 (3)	6.42 (5)	6.46 (6)
NB	6.48 (6)	6.58 (6)	6.52 (7)	6.28 (3)
LDA	6.68 (8)	6.84 (8)	6.50 (6)	6.36 (5)
LightGBM	5.34 (1)	4.94 (1)	5.18 (1)	4.98 (1)
LR	6.54 (7)	6.76 (7)	6.66 (8)	6.50 (7)
MLP	6.06 (3)	6.24 (3)	6.30 (4)	6.70 (10)
QDA	7.54 (12)	7.58 (12)	7.66 (12)	8.22 (12)
RF	6.08 (4)	6.26 (5)	5.72 (2)	5.88 (2)
Ridge	6.98 (10)	7.12 (11)	6.86 (9)	6.66 (8)
SGD	6.82 (9)	6.90 (9)	7.00 (10)	6.34 (4)

Distribution of Test-Set Ranks per Model

To better assess how each model performs across different types of evaluation criteria, Figure 4.5 displays the distribution of test-set ranks per model, separated into classification and utility metrics. This visualization allows us to identify not only overall model robustness but also whether strong statistical performance aligns with financial utility.

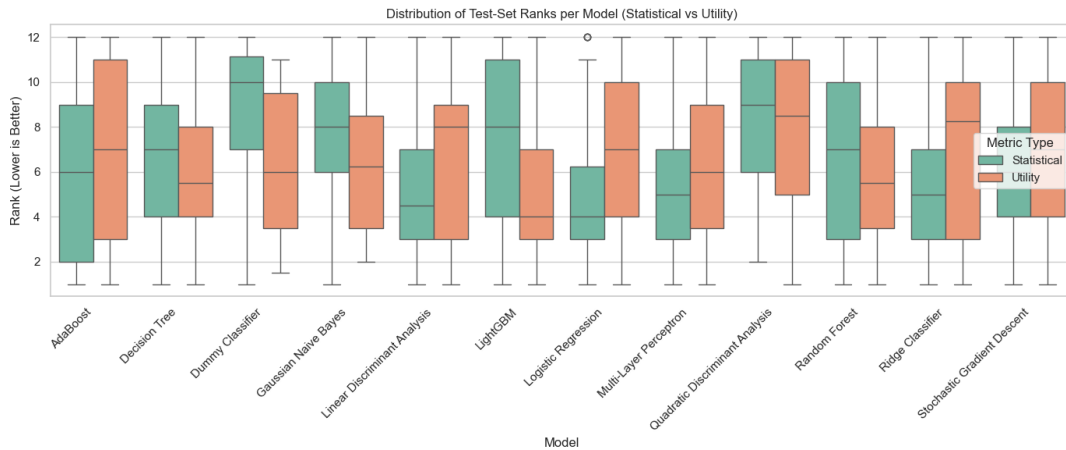


Figure 4.5: Distribution of test-set ranks per model, split by classification and utility metrics. Lower medians and narrower interquartile ranges indicate more consistent and high-performing models.

Tables 4.1 and 4.2, along with Figure 4.5, present model rankings under two evaluation lenses: traditional statistical performance and utility-based metrics. In Table 4.1, models such as Logistic Regression, Ridge Classifier, and MLP consistently achieve low ranks, indicating strong performance across metrics classification metrics.

In contrast, Table 4.2 shows that the highest-ranked models under utility metrics (LightGBM, Random Forest, and Decision Tree), are not those with the best classification scores. LightGBM, in particular, ranks first across all four utility-based metrics, despite finishing near the bottom of the classification rankings. This suggests that the relationship between classification accuracy and practical trading outcomes is not straightforward in this context.

In summary, the results reinforce the need for utility-aware model validation strategies. Sole reliance on traditional classification metrics like AUC or Accuracy can lead to suboptimal model selection for trading scenarios such as Binary options.

4.4 How does variation in binary options payout structures affect the relationship between accuracy and utility?

Binary options payouts play a critical role in model profitability. Because they fluctuate based on asset type, time of day, and volatility (Cofnas, 2016; TradingPedia, 2023), it is crucial to evaluate whether our findings generalize across different payout levels.

To answer the forth research question, we repeated the same correlation analysis under two extreme but realistic payout settings: 90% (favorable) and 60% (unfavorable).

Empirical Results Under Different Payout Regimes

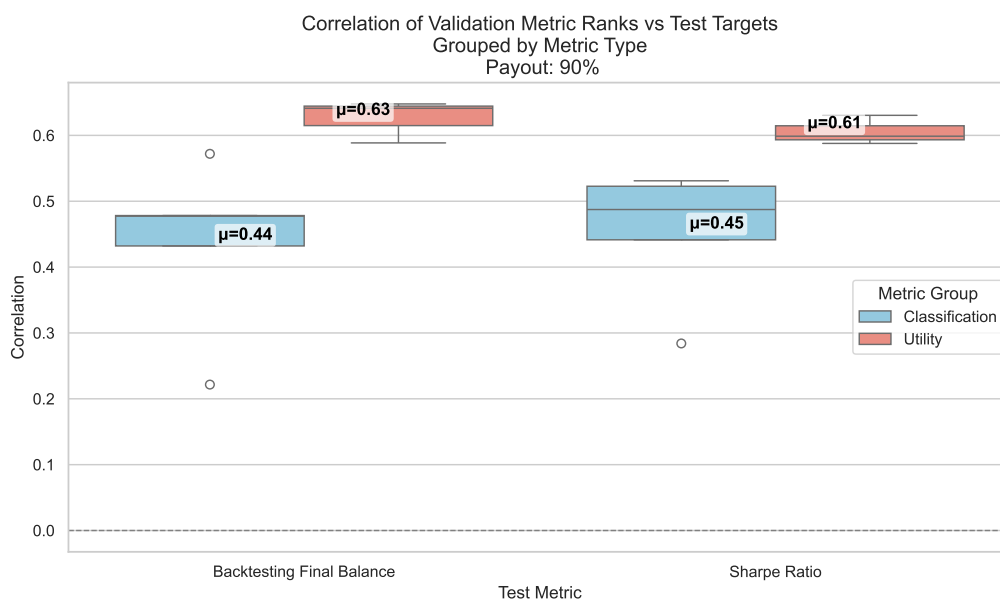


Figure 4.6: Correlation of validation metric rankings with test performance under a 90% payout structure. Utility metrics show high alignment with realized financial outcomes.

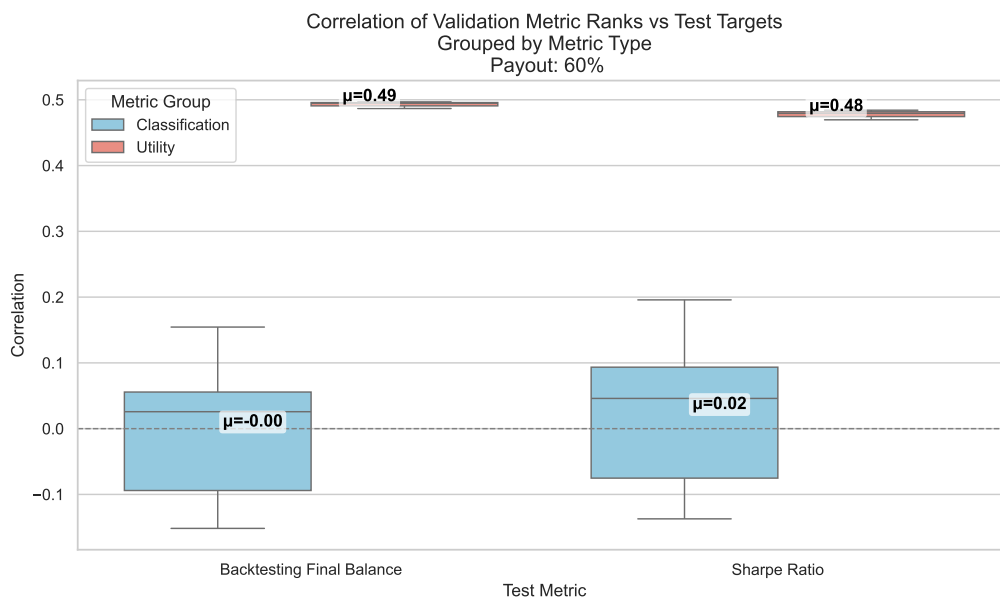


Figure 4.7: Correlation of validation metric rankings with test performance under a 60% payout structure. Correlations across all metric types decrease significantly.

At 90% payout Figure 4.6, utility metrics remained strong indicators of test-time financial performance (mean $\mu \approx 0.61$ – 0.63), while classification metrics offered modest but usable guidance (mean $\mu \approx 0.44$ – 0.45).

However, this relationship weakens considerably under the 60% payout structure Figure 4.7. Classification metrics showed no meaningful correlation with financial outcomes, averaging near-zero or even negative values (mean $\mu \approx 0.00$ – 0.02). Although utility metrics also experienced reduced performance it retained partial effectiveness (mean $\mu \approx 0.49$ – 0.48).

The combined results across all three payout settings (60%, 75%, 90%) indicate that:

- Utility metrics are more robust predictors of financial outcomes regardless of payout structure.
- Classification metrics are sensitive to payout volatility, and may only be useful under highly favorable market conditions.

These insights reinforce the importance of selecting models based on utility-driven validation criteria, especially in domains where market conditions can dramatically shift the economic implications of predictive accuracy.

Chapter 5

Conclusions

Forecasting in financial domains has traditionally emphasized the use of classification metrics such as Accuracy, F1 Score, and AUC as proxies for predictive quality (Niculescu-Mizil and Caruana, 2005; Fawcett, 2006; Hand, 2012). However, as highlighted in this study, the assumption that such metrics adequately capture the economic utility of a model is increasingly questioned, particularly in contexts characterized by asymmetric or fat-tailed payoffs (Taleb, 2020). This thesis aimed to explore this disconnect in the specific setting of binary options trading, where outcomes are strictly defined: a trader either receives a fixed payout or loses the entire amount invested (Cofnas, 2016).

5.1 Main Results

By framing short-term directional price forecasting as a binary classification task and implementing a backtesting framework that mimics real-world trading platforms (e.g., IQ Option, Nadex) (Option, 2025b), this study enabled a dual-layered evaluation of model performance. On the one hand, traditional classification metrics assessed predictive performance, and on the other hand, utility-based metrics provided a more grounded view of a model's real-world applicability and financial impact.

The results reveal that models with superior classification performance do not consistently yield higher financial returns. Correlation studies across various payout structures (60%, 75%, and 90%) demonstrate that classification validation metrics exhibit weak and unstable alignment with backtesting profitability. In contrast, utility-based metrics, particularly those incorporating risk-adjusted returns, show stronger and more consistent correlation with financial outcomes (see Figures 4.3, 4.4, 4.6, 4.7).

Our analysis also reveals a varying relative performance of different models across different types of metrics (Section 4.3). Models that excel in traditional classification metrics often fail to deliver a corresponding financial performance relative to other models, and vice versa. This pattern is exemplified by `LightGBM`, which achieved the best utility performance while ranking poorly in classification metrics.

Furthermore, rank-delta analyses (Figures 4.1, 4.2) reinforce these findings. Improvements in classification metrics such as Accuracy do not reliably coincide with increases in financial performance, in many cases, models with better classification metrics underperform financially. Conversely, improvements in Sharpe Ratio often accompany improvements in backtesting financial outcomes, suggesting these are more reliable signals for model selection in economic contexts.

5.2 Limitations and Future Work

While the findings are robust across multiple assets and model classes, certain limitations warrant consideration. The use of a fixed 75% payout structure, while methodologically necessary to ensure consistent comparison across models and assets, represents a simplification of real-world conditions where payouts typically vary by asset, time of day, and market volatility (TradingPe-dia, 2023). Additionally, the backtesting framework assumes ideal execution conditions (e.g., no slippage, perfect timing), which may overstate actual profitability. Finally, reliance on technical indicators assumes that price patterns exhibit an exploitable structure, a premise that may not always hold in efficient markets.

Future work may extend this analysis by incorporating adversarial conditions such as latency or slippage. Another promising direction is the use of reinforcement learning to directly optimize for utility-based objectives, bypassing traditional supervised learning bottlenecks. Finally, expanding this framework to multiclass or continuous payout options (e.g., digital options with sliding payoff structures) would generalize its applicability and test the scalability of the proposed evaluation paradigm.

5.3 Concluding Remarks

This thesis supports the claim that forecasting accuracy and economic utility are not equivalent, especially in constrained, high-stakes environments like binary options trading. Our findings provide further empirical support for the argument that forecasting accuracy metrics alone are not a sufficient proxy for real-world payoffs. This observation corroborates the conclusions from previous work in various domains (Taleb, 2020; Yardley and Petropoulos, 2021). In alignment with Yardley and Petropoulos (2021), we also advocate for the use of utility evaluation metrics when evaluating forecasting models whenever possible.

References

- Perceval Beja-Battais. Overview of adaboost: Reconciling its views to better understand its dynamics. *arXiv preprint arXiv:2310.18323*, 2023. URL <https://arxiv.org/abs/2310.18323>.
- Leo Breiman. Random forests. *Machine learning*, 45(1):5–32, 2001. URL <https://link.springer.com/article/10.1023/A:1010933404324>.
- G. W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78:1–3, 1950. URL [https://doi.org/10.1175/1520-0493\(1950\)078%3C0001:VOFEIT%3E2.0.CO;2](https://doi.org/10.1175/1520-0493(1950)078%3C0001:VOFEIT%3E2.0.CO;2).
- Trading Center. Trading binary options, 2025. URL <https://tradingcenter.org/index.php/trade/forex/182-trading-binary-options>. Accessed: 2025-06-26.
- CFTC and SEC. Cftc/sec investor alert: Binary options and fraud. *Commodity Futures Trading Commission*, 2018. URL https://www.cftc.gov/LearnAndProtect/AdvisoriesAndArticles/fraudadv_binaryoptions.html.
- Abe Cofnas. *Trading Binary Options: Strategies and Tactics: Second Edition*. Bloomberg Financial, 07 2016. ISBN 9781119194170. doi: 10.1002/9781119270027.
- David R Cox. The regression analysis of binary sequences. *Journal of the Royal Statistical Society: Series B (Methodological)*, 20(2):215–242, 1958. URL <https://rss.onlinelibrary.wiley.com/doi/abs/10.1111/j.2517-6161.1958.tb00292.x>.
- Francis Dama and Christine Sinoquet. Time series analysis and modeling to forecast: A survey. *arXiv preprint arXiv:2104.00164*, 2021. URL <https://arxiv.org/abs/2104.00164>.
- Tom Fawcett. An introduction to roc analysis. *Pattern Recognition Letters*, 27(8):861–874, 2006. ISSN 0167-8655. doi: <https://doi.org/10.1016/j.patrec.2005.10.010>. URL <https://www.sciencedirect.com/science/article/pii/S016786550500303X>. ROC Analysis in Pattern Recognition.
- Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273, 2020. URL <https://doi.org/10.1093/rfs/hhaa009>.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1321–1330, 2017. URL <https://doi.org/10.48550/arXiv.1706.04599>.

- David J. Hand. Assessing the performance of classification methods. *International Statistical Review*, 80(3):400–414, 2012. doi: <https://doi.org/10.1111/j.1751-5823.2012.00183.x>. URL <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1751-5823.2012.00183.x>.
- Arthur E Hoerl and Robert W Kennard. Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67, 1970. URL <https://www.tandfonline.com/doi/abs/10.1080/00401706.1970.10488634>.
- Rob J Hyndman and George Athanasopoulos. *Forecasting: principles and practice*. OTexts, 2018. URL <https://otexts.com/fpp3/index.html>.
- Sakina Ichani and Agatha Rinta Suhardi. The effect of return on equity (roe) and return on investment (roi) on trading volume. *Procedia - Social and Behavioral Sciences*, 211:896–902, 2015. ISSN 1877-0428. doi: <https://doi.org/10.1016/j.sbspro.2015.11.118>. URL <https://www.sciencedirect.com/science/article/pii/S1877042815054580>. 2nd Global Conference on Business and Social Sciences (GCBSS-2015) on “Multidisciplinary Perspectives on Management and Society”, 17- 18 September, 2015, Bali, Indonesia.
- Investopedia. Cumulative return: Definition, calculation, and example. *Investopedia*, 2020. URL <https://www.investopedia.com/terms/c/cumulativereturn.asp>.
- Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. Lightgbm: A highly efficient gradient boosting decision tree. In *Advances in neural information processing systems*, pages 3146–3154, 2017. URL https://papers.nips.cc/paper_files/paper/2017/file/6449f44a102fde848669bdd9eb6b76fa-Paper.pdf.
- J. L. Kelly. A new interpretation of information rate. *Bell System Technical Journal*, 35(4):917–926, 1956. URL https://www.princeton.edu/~wbialek/rome/refs/kelly_56.pdf. Reproduced with permission of AT&T.
- Preetum Nakkiran, Gal Kaplun, Dimitris Kalimeris, Tristan Yang, Benjamin L Edelman, Fred Zhang, and Boaz Barak. Sgd on neural networks learns functions of increasing complexity. *arXiv preprint arXiv:1905.11604*, 2019. URL <https://arxiv.org/abs/1905.11604>.
- NASAA. Informed investor advisory: Binary options. *North American Securities Administrators Association*, 2017. URL <https://www.nasaa.org/43192/informed-investor-advisory-binary-options/>.
- Alexandru Niculescu-Mizil and Rich Caruana. Predicting good probabilities with supervised learning. In *Proceedings of the 22nd International Conference on Machine Learning (ICML)*, pages 625–632, 01 2005. doi: 10.1145/1102351.1102430.
- IQ Option. Trading binary options on iq option: Detailed guide. *IQ Option Broker Official Blog*, 2025a. URL <https://blog.iqoption.com/en/basics-of-binary-options-trading-explained>. Accessed: 28 April 2025.
- IQ Option. How to trade digital options: Are they better than binary options? *IQ Option Broker Official Blog*, 2025b. URL <https://blog.iqoption.com/en/how-to-trade-digital-options-an-advanced-version-of-binary-options/#:~:text=3,of%20the%20nearest%20strike%20price>. Accessed: 28 April 2025.

- IQ Option. Iq option for beginners: a basic guide. *IQ Option Broker Official Blog*, 2025c. URL <https://blog.iqoption.com/en/how-to-trade-a-beginners-guide-to-trading/>. Accessed: 28 April 2025.
- Polygon.io. Where does polygon source ticker news from?, 2025. URL <https://polygon.io/knowledge-base/article/where-does-polygon-source-ticker-news-from>. Accessed: 2025-05-21.
- Marius-Constantin Popescu, Valentina Balas, Liliana Perescu-Popescu, and Nikos Mastorakis. Multilayer perceptron and neural networks. *WSEAS Transactions on Circuits and Systems*, 8, 07 2009.
- Foster Provost and Tom Fawcett. Data science and its relationship to big data and data-driven decision making. *Big Data*, 1(1):51–59, 2013. doi: 10.1089/big.2013.1508. URL <https://doi.org/10.1089/big.2013.1508>. PMID: 27447038.
- Irina Rish. An empirical study of the naive bayes classifier. In *IJCAI 2001 Workshop on Empirical Methods in Artificial Intelligence*, pages 41–46. IBM New York, 2001. URL <https://faculty.cc.gatech.edu/~isbell/reading/papers/Rish.pdf>.
- Lior Rokach and Oded Maimon. *Decision Trees*, volume 6, pages 165–192. Research Gate, 01 2005. doi: 10.1007/0-387-25465-X_9.
- scikit-learn. *sklearn.dummy.DummyClassifier — scikit-learn 1.6.1 documentation*. scikit-learn, 2025. URL <https://scikit-learn.org/stable/modules/generated/sklearn.dummy.DummyClassifier.html>. Accessed: 2025-05-15.
- Australian Securities and Investments Commission. Extension of the binary options product intervention order, 2022. URL <https://download.asic.gov.au/media/mjkhommx/cp362-published-26-may-2022.pdf>.
- U.S. Securities and Exchange Commission. Binary options. *Investor.gov*, 2025. URL <https://www.investor.gov/introduction-investing/investing-basics/glossary/binary-options>.
- William F Sharpe. Mutual fund performance. *The Journal of Business*, 39(1):119–138, 1966. URL <https://finance.martinsewell.com/fund-performance/Sharpe1966.pdf>.
- Gurjeet Singh. Machine learning models in stock market prediction. *arXiv Preprint*, abs/2202.09359, 2022. URL <https://arxiv.org/abs/2202.09359>.
- Marina Sokolova and Guy Lapalme. A systematic analysis of performance measures for classification tasks. *Information Processing & Management*, 45(4):427–437, 2009. ISSN 0306-4573. doi: <https://doi.org/10.1016/j.ipm.2009.03.002>. URL <https://www.sciencedirect.com/science/article/pii/S0306457309000259>.
- Frank A Sortino and Lee N Price. Performance measurement in a downside risk framework. *The Journal of Investing*, 1994. URL <https://www.pm-research.com/content/iijinvest/3/3/59>.
- Nassim Nicholas Taleb. On the statistical differences between binary forecasts and real-world payoffs. *International Journal of Forecasting*, 36(4):1228–1240, 2020. ISSN 0169-2070. doi: <https://doi.org/10.1016/j.ijforecast.2019.12.004>. URL <https://www.sciencedirect.com/science/article/pii/S0169207019302699>.

- Alaa Tharwat, Tarek Gaber, Abdelhameed Ibrahim, and Aboul Ella Hassanien. Linear discriminant analysis: A detailed tutorial. *Ai Communications*, 30:169–190,, 05 2017. doi: 10.3233/AIC-170729.
- Olymp Trade. Digital options with olymp trade – online trading made simple. *Entrepreneur*, 2018. URL <https://www.entrepreneur.com/en-za/technology/digital-options-with-olymp-trade-online-trading-made/330572>.
- TradingPedia. Payouts in binary options. *TradingPedia Binary Options Academy*, 2023. URL <https://www.tradingpedia.com/binary-options-academy/payout/>.
- Sudhanshu Sekhar Tripathy and Bichitrananda Behera. Performance evaluation of machine learning algorithms for intrusion detection system. *arXiv Preprint*, abs/2310.00594, 2023. URL <https://arxiv.org/abs/2310.00594>.
- U.S. Securities and Exchange Commission. Self-regulatory organizations; american stock exchange llc; order approving proposed rule change and amendment no. 1 thereto relating to the listing and trading of notes linked to the performance of the VIX index, oct 2008. URL <https://www.sec.gov/files/rules/sro/amex/2008/34-58272.pdf>.
- Johannes Hendrik Venter and Pieter Juriaan De Jongh. Trading binary options using expected profit and loss metrics. *Risks*, 10(11), 2022. doi: 10.3390/risks10110212. URL <https://doi.org/10.3390/risks10110212>. Special Issue: New Advance of Risk Management Models.
- Ralph Vince. *The Mathematics of Money Management: Risk Analysis Techniques for Traders*. John Wiley & Sons, New York, 1992. ISBN 978-0471547380. URL <http://dl.fxfl.com/books/english/MathematicsMoneyManagement.pdf>.
- Ruiyang Wu and Ning Hao. Quadratic discriminant analysis by projection. *arXiv preprint arXiv:2108.09005*, 2021. URL <https://arxiv.org/abs/2108.09005>.
- Piotr Wójcik. Money management methods in trading and investing. *ResearchGate*, 2017. URL https://www.researchgate.net/publication/315743262_Money_management_methods_in_trading_and_investing.
- L. Yardley and F. Petropoulos. Beyond error measures to the utility and cost of the forecasts. *Foresight: the International Journal of Applied Forecasting*, 63: 36–45, 2021. URL https://purehost.bath.ac.uk/ws/portalfiles/portal/226468715/Beyond_error_measures_20210901.pdf. CC BY-NC-ND.
- Qizhi Zheng, Ayang Zhao, Xinzhu Wang, Yanhong Bai, Zikun Wang, Xiuying Wang, Xianzhang Zeng, and Guanghui Dong. Machine learning algorithms to predict stroke in china based on causal inference of time series analysis. *arXiv Preprint*, abs/2503.14512, 2025. URL <https://arxiv.org/abs/2503.14512>.